



SENTINEL

Protecting the power grid from hacked EV chargers

Cyberthorn · Cybersecurity & Critical Infrastructure Protection · [Team name] · [Members] · [University]



Try the live demo

rithikhc.github.io/Cyberthorn

1 Introduction

Everything runs on the grid

Hospitals, water and desalination, cooling, metro and airports, banks, telecom and data centres all stop without power, and so do factories and cold storage. A grid failure hits a whole economy at once, so it is critical national infrastructure.

External attackers have already shown it can be done. In 2015, hackers opened breakers at 30 substations in Ukraine and cut power to about 225,000 customers.

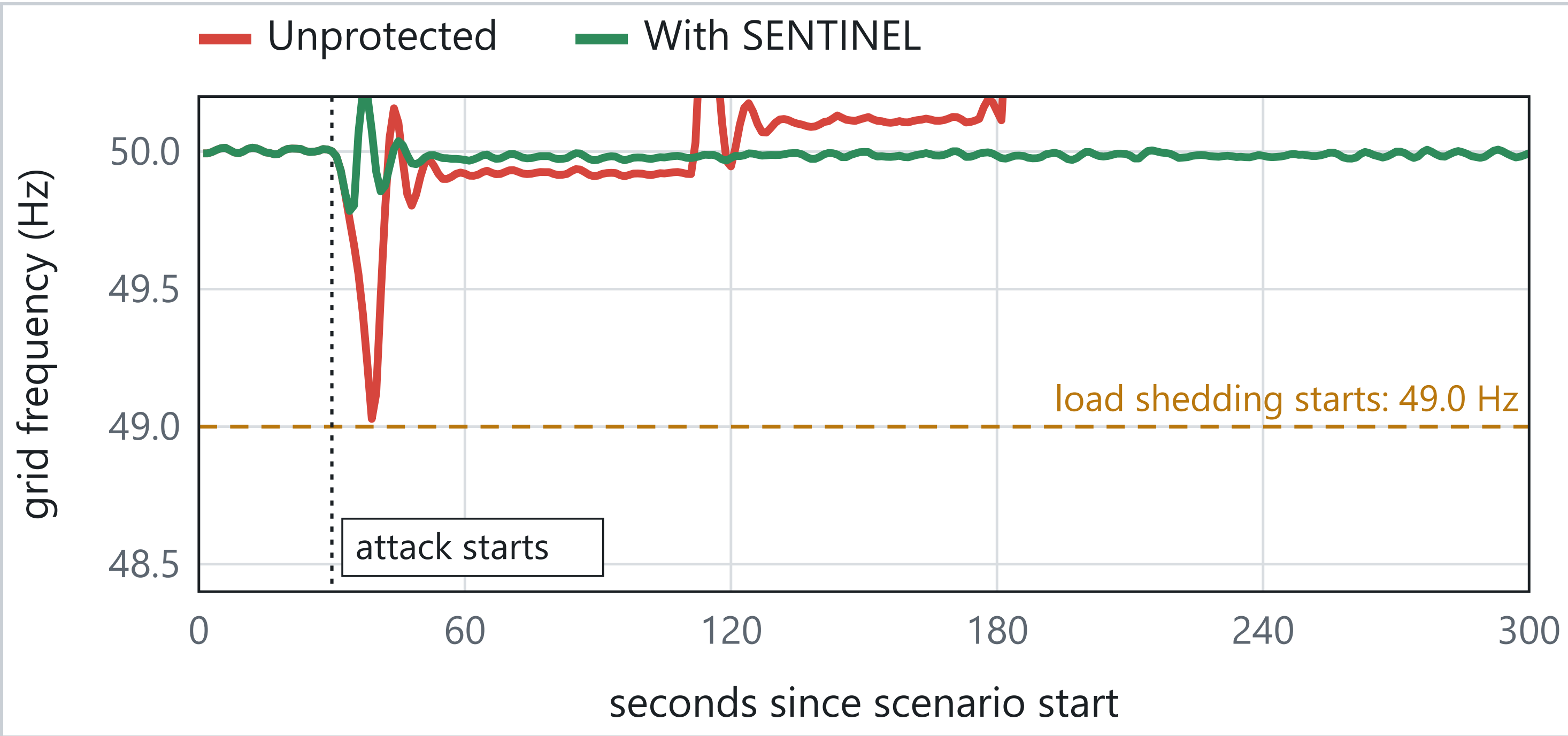
EVs are going mainstream, and so are chargers

- Over **20%** of new cars sold worldwide in 2024 were electric; the IEA expects over **40%** by 2030.
- UAE policy targets EVs at about **10%** of the fleet by 2030 and **50%** by 2050.
- Dubai had **47,944** EVs and about **1,860** public charge points at the end of 2025.
- ADNOC and TAQA aim for **70,000** charge points in Abu Dhabi by 2030.

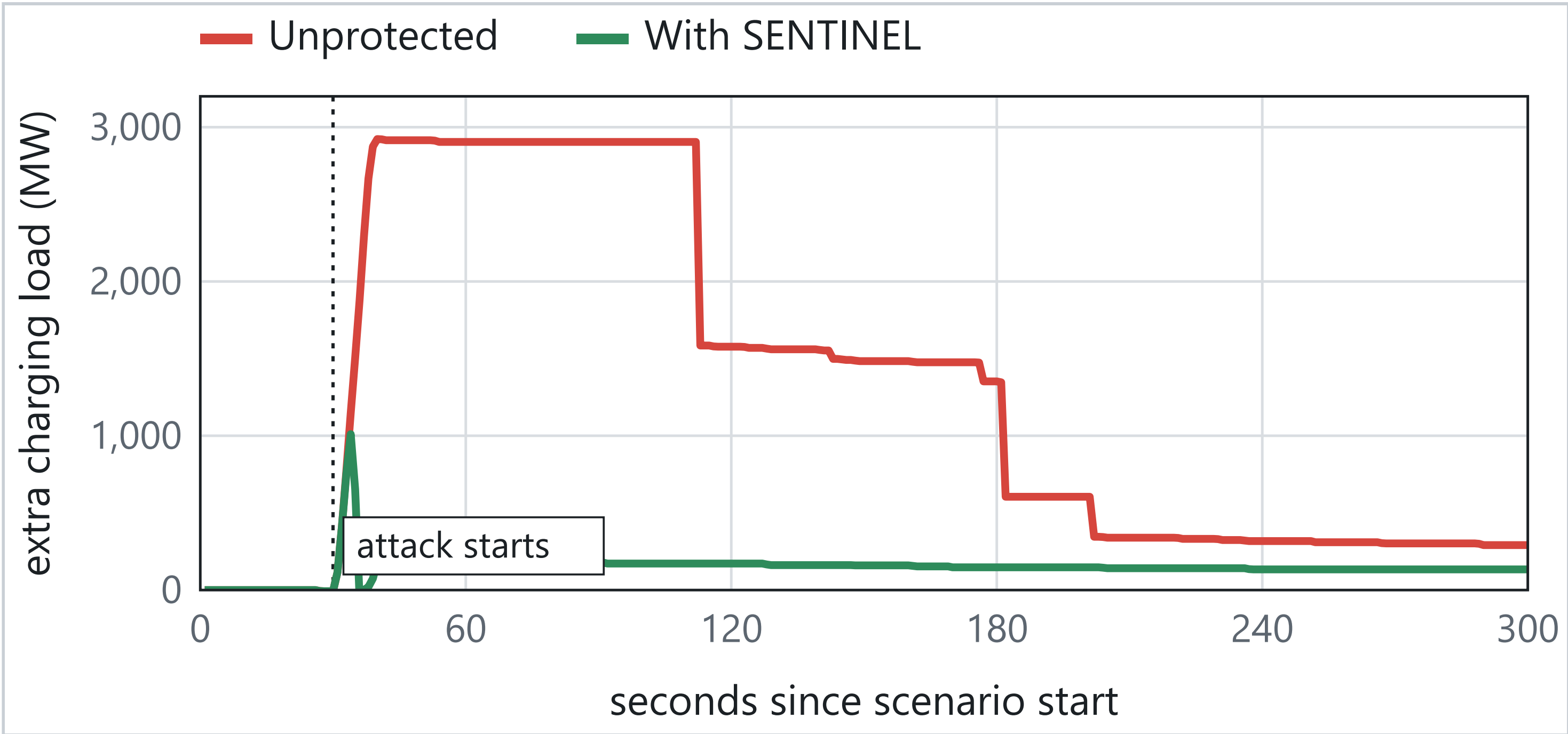
~34 GW UAE peak electricity demand, 2024	225,000 customers cut off in the 2015 Ukraine attack
>20% of new cars sold worldwide are electric	70,000 charge points targeted in Abu Dhabi by 2030

4 Results: the same attack, run twice

Grid frequency: nationwide attack, 60% of chargers hacked

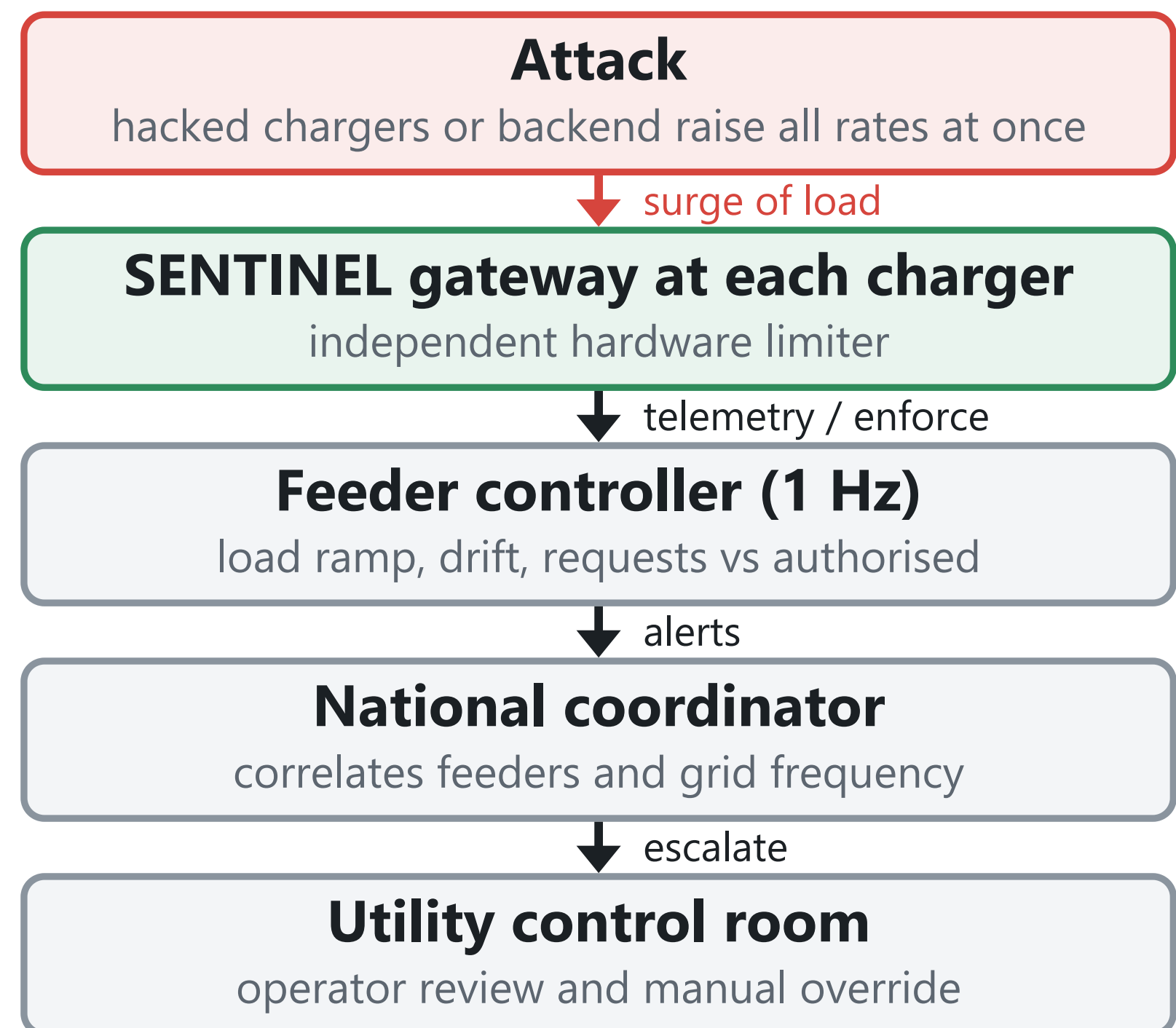


Extra charging load on the grid



SCENARIO (2040, OVERNIGHT)	WITHOUT SENTINEL	WITH SENTINEL
Nationwide, 60% of chargers hacked	2,131 MW of customers shed, 5 breakers tripped, 48.94 Hz	No load shed, no trips, 49.77 Hz. Throttle in 3 s
Local attack on Dubai's EV-dense districts	Feeder breaker trips	No trip
Slow ramp, charging backend compromised	2,131 MW shed	No load shed, throttle in 9 s
Legitimate off-peak tariff surge	n/a	No false alarm, nothing quarantined

5 System



6 What the model shows

~600,000
hacked chargers (about 40% of those installed) start load shedding nationwide by 2040

~1 in 4
chargers on one dense Dubai feeder can trip its breaker

~98%
of attackable overnight load comes from home chargers

3 s
to start throttling (9 s for a slow ramp)

- Daytime attacks are far weaker: cars are not plugged in.
- Overdriving past the battery limit makes cars cut off, so the attack defeats itself.
- Limit:** with control of only 30% of chargers, a 2050 attack is only partly contained (10,656 MW shed falls to 4,262 MW). Coverage matters.

7 Impact & next steps

Why it matters

- Fits the UAE National Cybersecurity Strategy: critical-infrastructure protection, detection and response, secure IoT.
- Scales as chargers multiply, because it watches the feeder, not each device.

Next steps

- ESP32 hardware gateway: current sensing, independent pilot-signal limiter, contactor, signed commands.
- Validate on real utility data and a hardware-in-the-loop bench.
- Integrate OCPP, IEC 61850, DNP3 and PMU data.